

LAB MANUAL

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1. Objective

- 1.1. The objective of this lab is to create a java application and run on the emulator.
- 1.2. Build and install and Debug an android application.
- 1.3. To Explore the structure of the APK(Android Package Kit).

2. Prerequisites

Operating System	Ubuntu 20.04 / Windows 10
Hardware	● CPU: x86_64 CPU architecture ● RAM: 8 GB RAM or more ● Disk Space: 8 GB of available disk space minimum (IDE + Android SDK + Android Emulator)
Tools Required	Not Required

3. Problem Statement

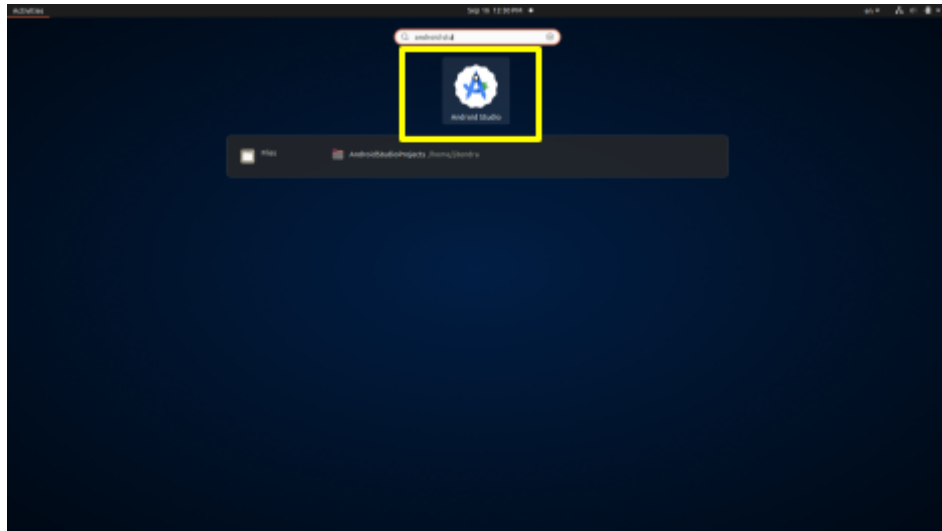
To develop a basic Java-based Android application using Android Studio, build and install the application on an Android Virtual Device (AVD), debug the application, and explore the internal structure and contents of the generated APK (Android Package Kit) to understand the fundamental components of an Android application.

4. Summary

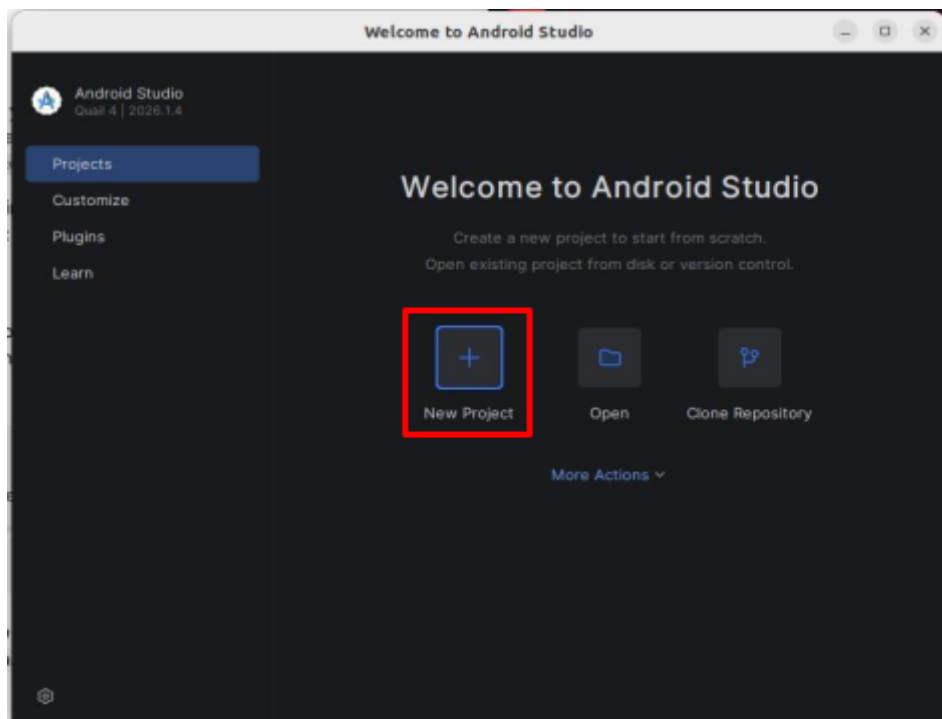
By implementing the steps from this module, we will be able to successfully develop a Java-based Android application using Android Studio, build, install, and debug the application on an Android Virtual Device (AVD), and explore the structure and contents of the generated APK file.

5. Create a Java Application :

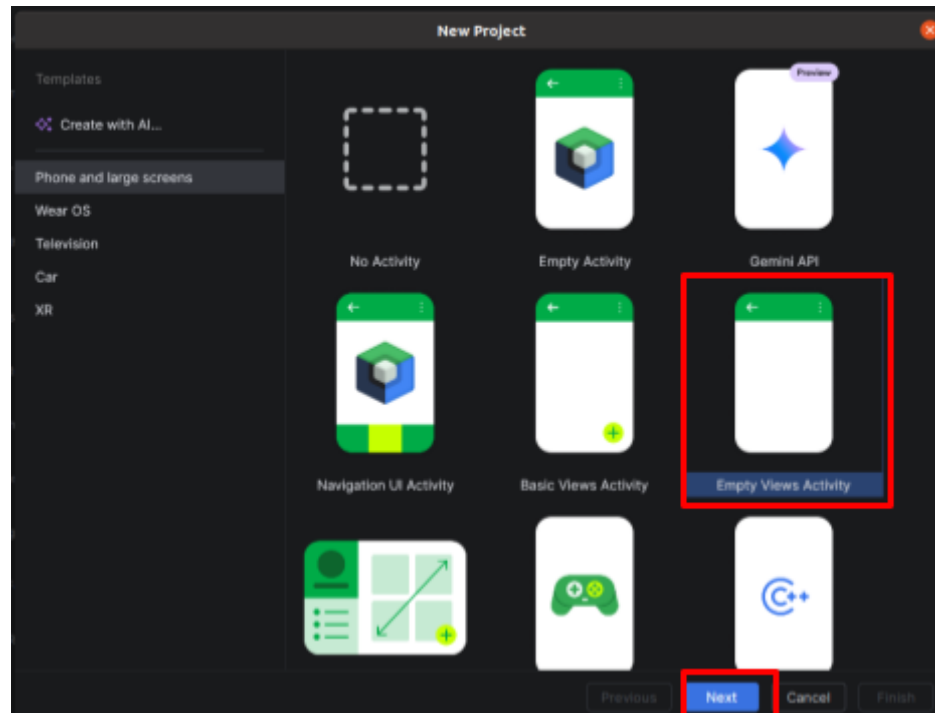
5.1. Open Android Studio



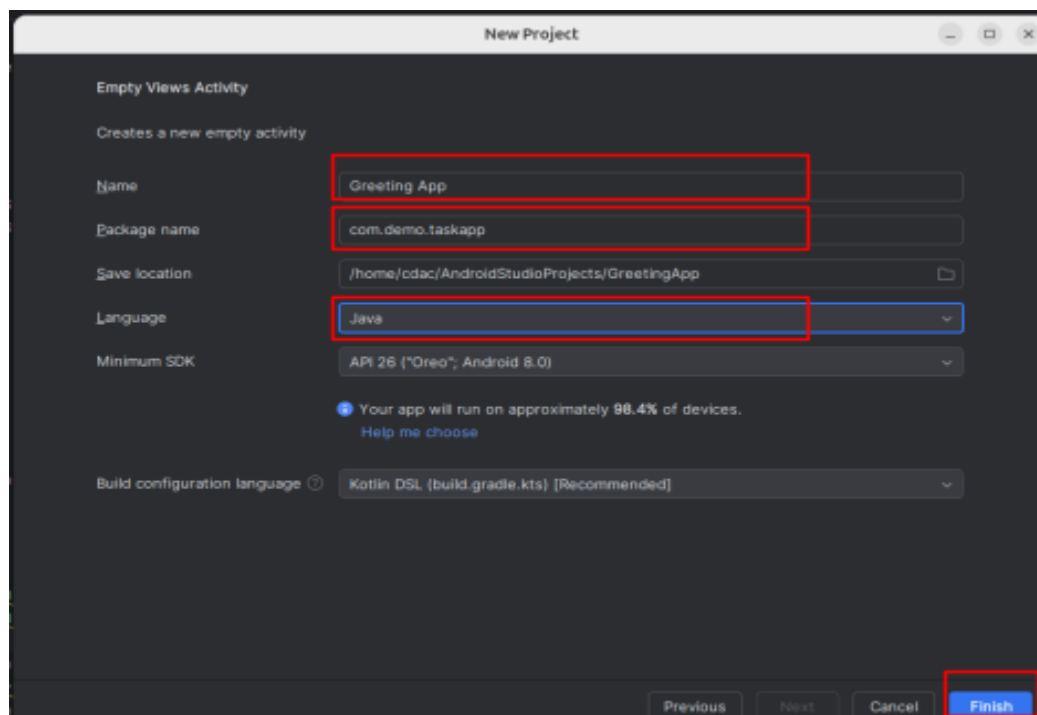
5.2. Click On New Project



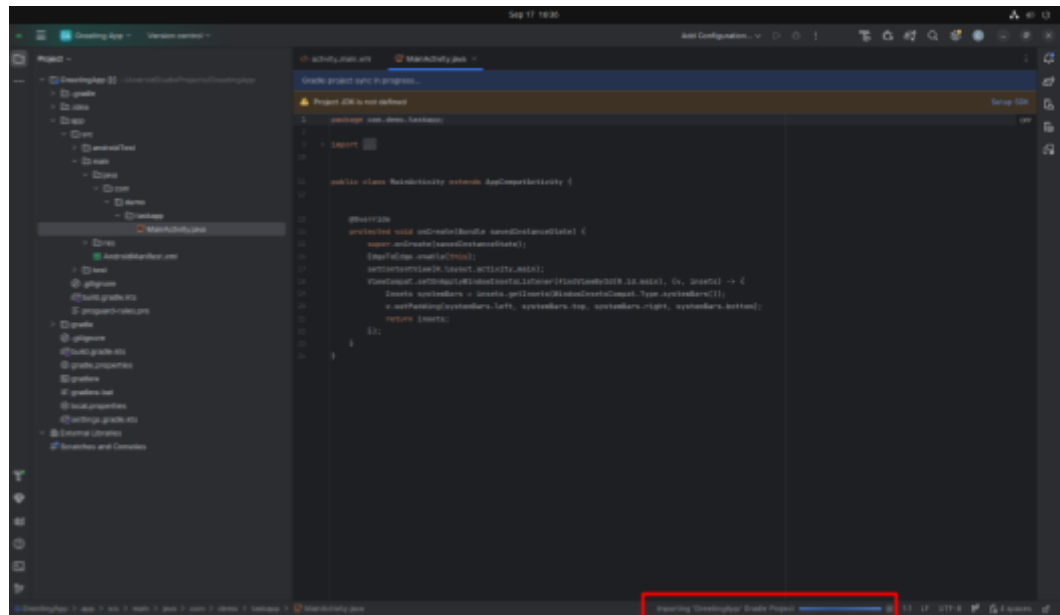
5.3. Select **Empty Views Activity** and click on **Next**.



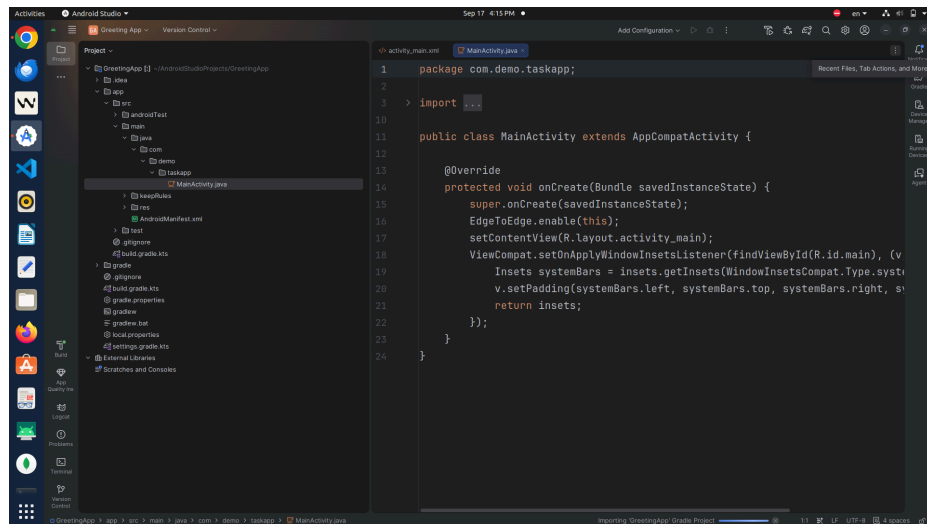
5.4. **Configure the Project : Name , Package Name , Minimum SDK.** also change the **Language** to **Java** if it is in **Kotlin** and Leave **Save Location** and build configuration language with default values and then Click on **Finish**.



5.5. Gradle Sync is in progress wait for sometime.

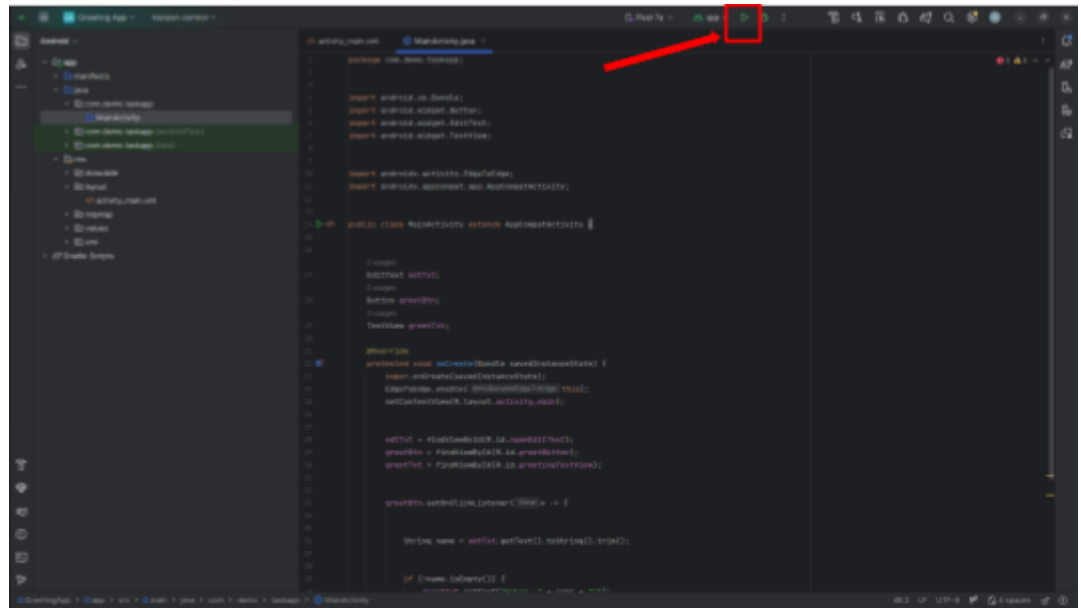


5.6. After Successful Gradle Sync we will have this interface on the screen.

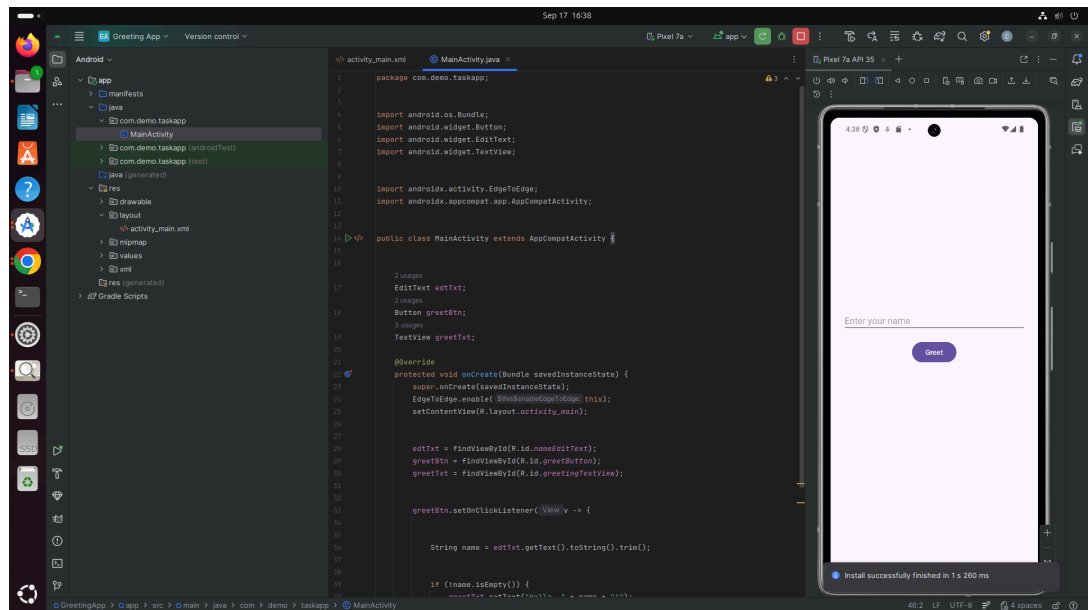


5.7. Change the code in the main activity and edit the xml files as required. After writing the entire code click on the green play icon on top. it will launch the

already installed emulator and our application will run.

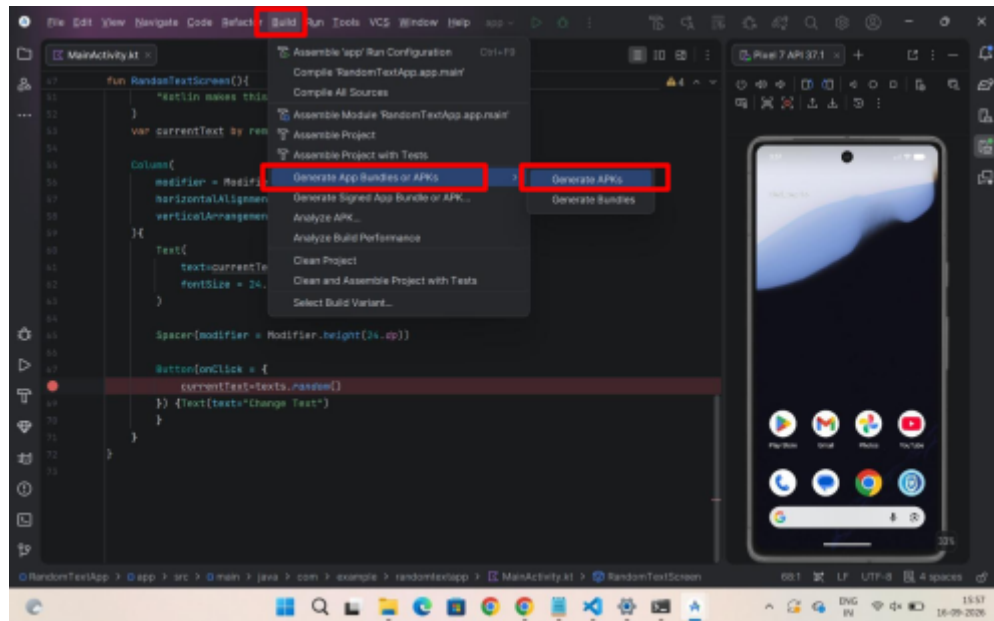


- 5.8. Emulator will be launched and our application will be running on the installed emulator.

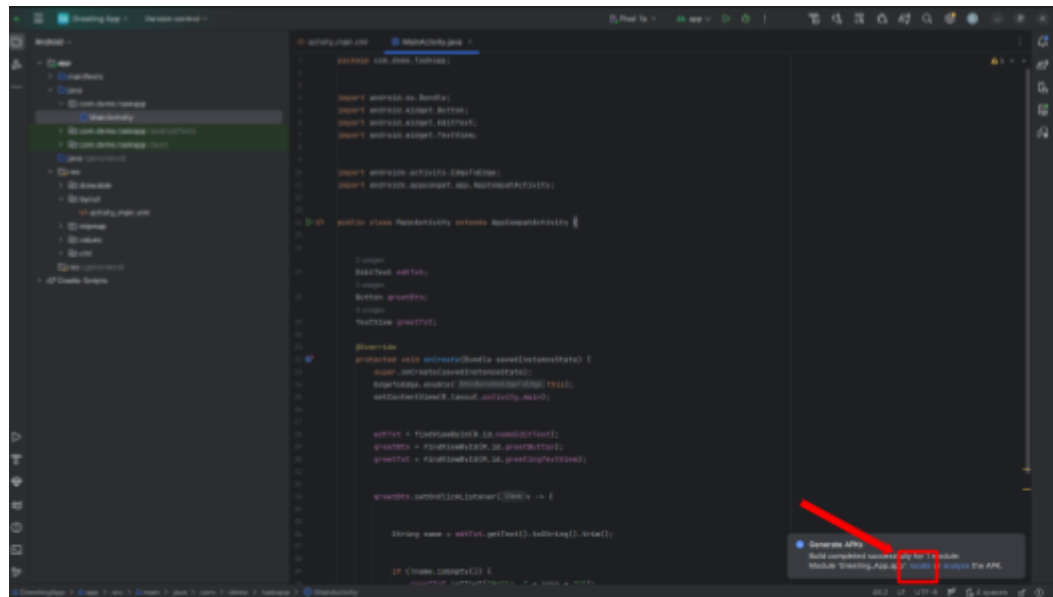


6. Build, Install And Debug APK

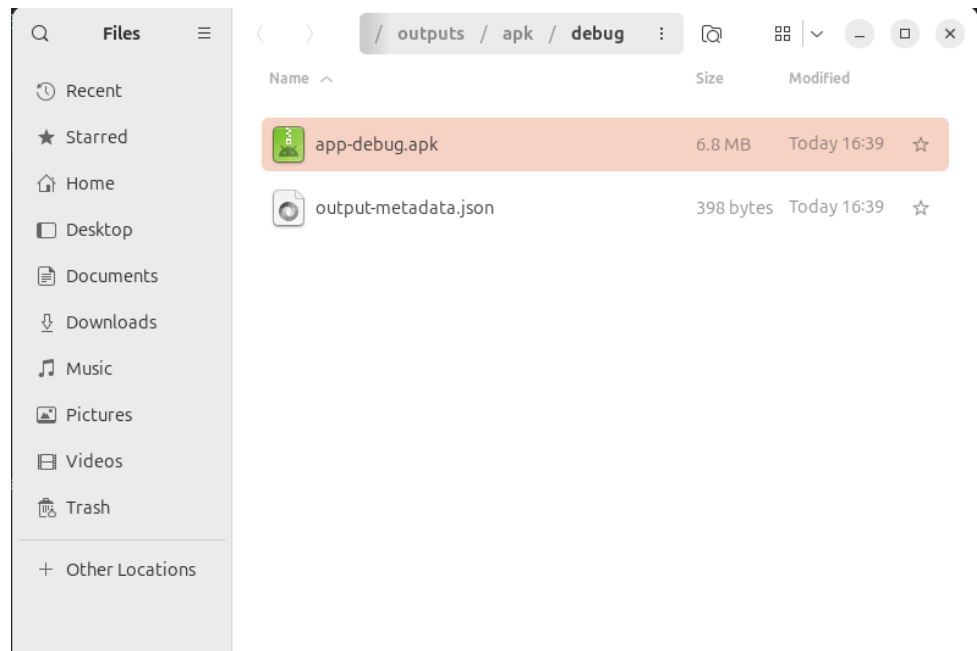
- 6.1. To Build an APK Click on Build at the top menu and click on Generate App Bundles or APK's and click on Generate APK's.



- 6.2. After the completion of the running of the gradle at the bottom right of the screen a pop up with the location of the generated apk. Click on Locate.



6.3. We will be navigated to the computer's file system where the apk is located.

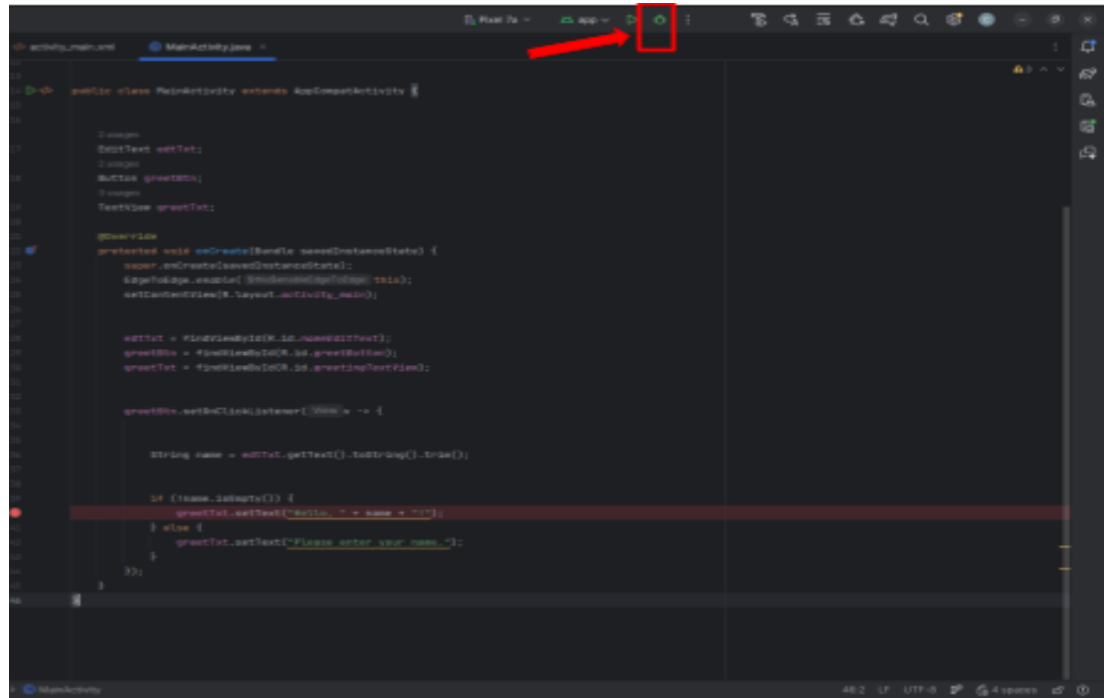


6.4. **Debugging of APK :** To debug an APK first create a breakpoint by clicking on the code line number, a red background will be applied to that line that is the break point.

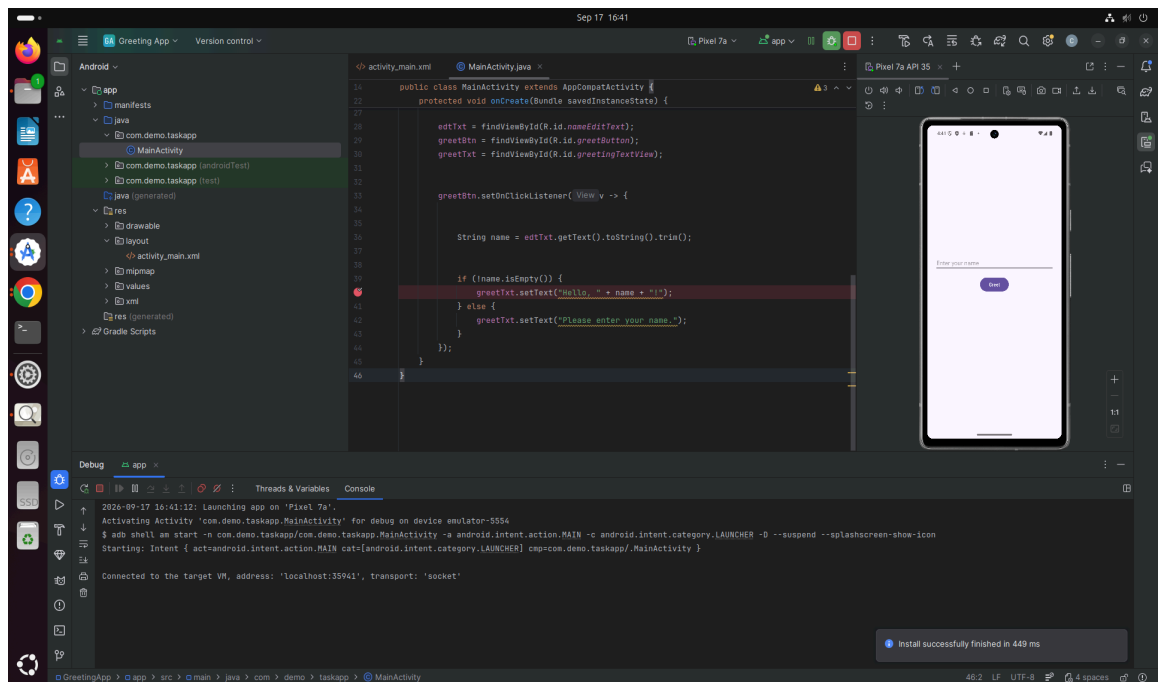


```
31
32
33     greetBtn.setOnClickListener(View v -> {
34
35         String name = editText.getText().toString().trim();
36
37
38         if (!name.isEmpty()) {
39             greetTxt.setText("Hello, " + name + "!");
40         } else {
41             greetTxt.setText("Please enter your name.");
42         }
43     });
44 }
45
46 }
```

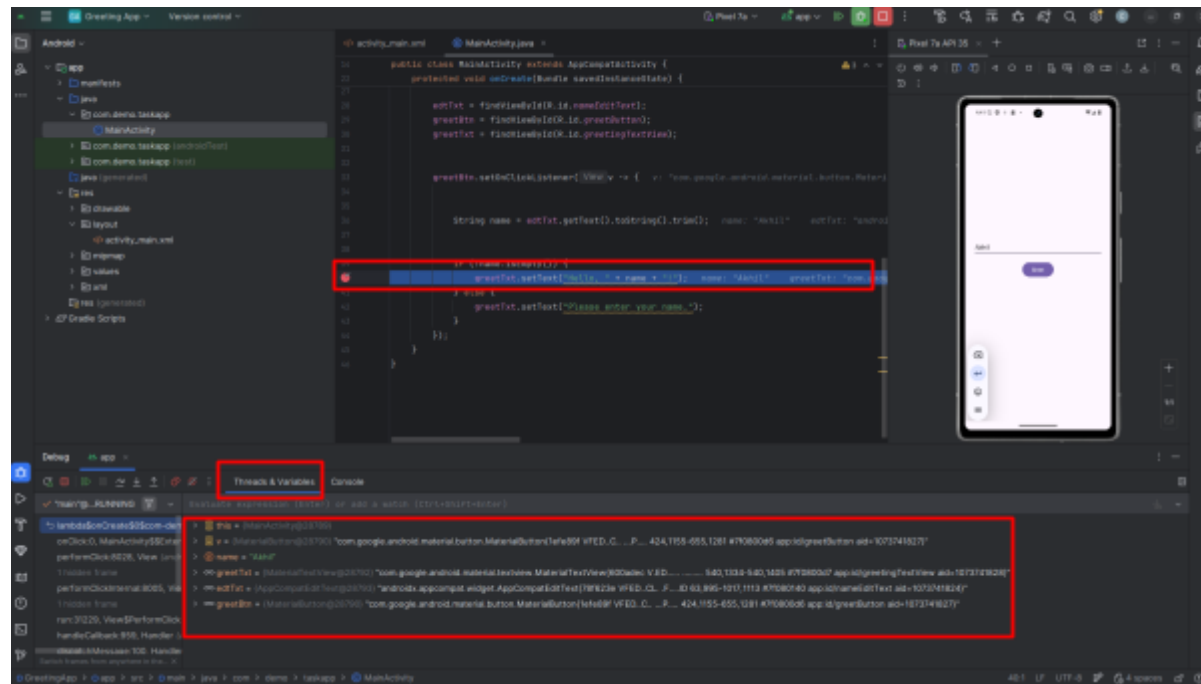
- 6.5. Now instead of clicking on the play button to run the application , now click on the bug icon beside it to run



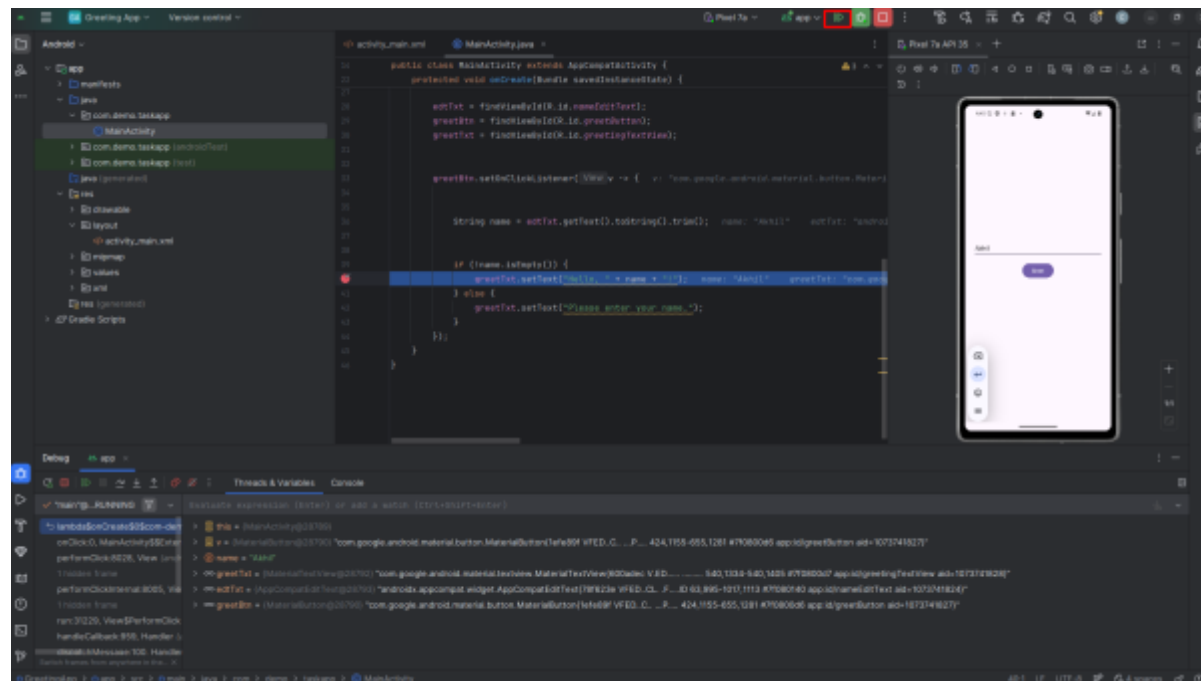
- 6.6. After successfully attaching the debugger this type of interface will be displayed on the screen.



- 6.7. Now click on Thread & Variables to see and debug how the variables state and value is changing. When it's the time for the statement we marked as debugging to be executed the application will halt it's execution and the selected debugging statement will be highlighted.

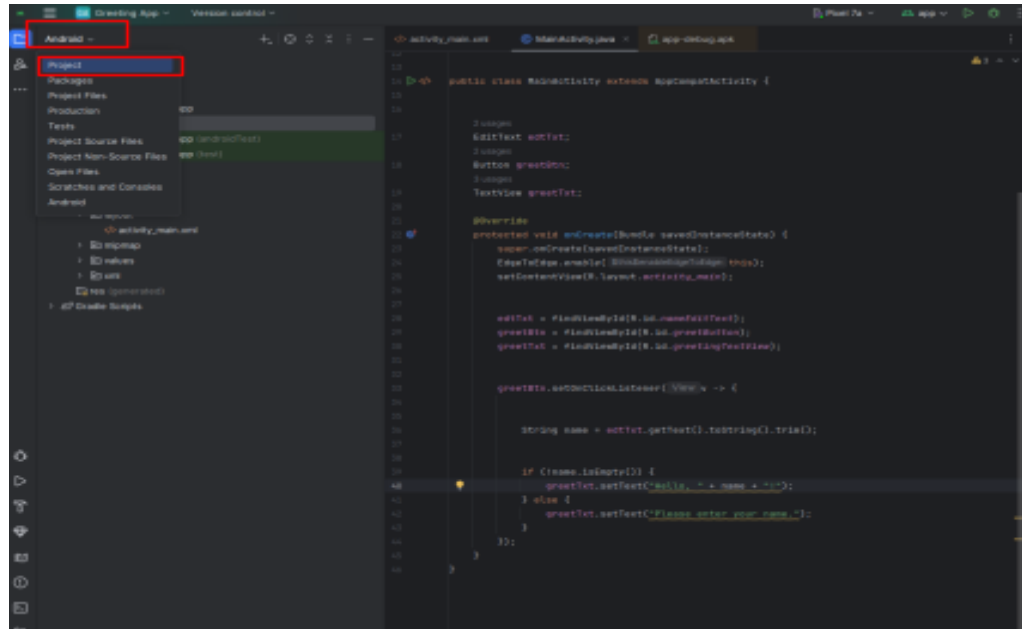


- 6.8. Now click on the Resume App Button which is left of the debug button to continue the execution of the application.

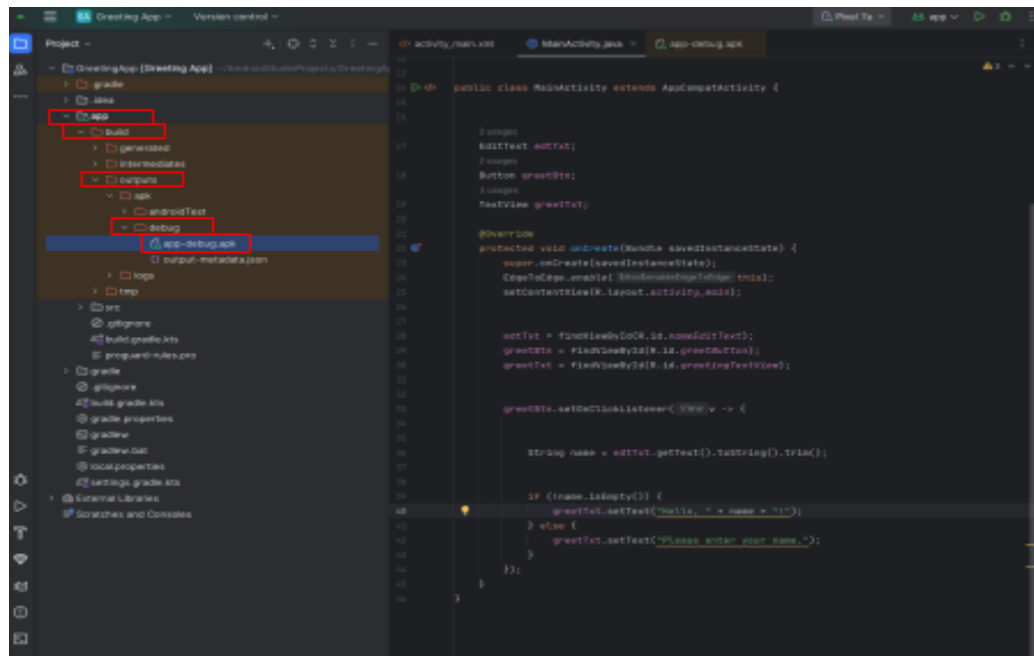


7. Explore APK Structure

- 7.1. After Successful generation of the APK using the Build Generate APK's(Which we have performed before in this manual). Change the view from the Android to Project on the top left of the screen.



- 7.2. In the File Structure navigate to app -> build->outputs -> apk->debug->app-debug.apk



- [illegible]

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